

Advanced Topics in Communication Systems

Lecture Notes

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Lecture 1

Notations

- x : Scalar
- $\mathbf{x}$: Vector
- $\mathbf{X}$: Matrix
- $\mathbb{E}[\cdot]$: Expectation Operator
- $\text{Var}(\cdot)$: Variance Operator
- $\text{Cov}(\cdot, \cdot)$: Covariance Operator
- $\mathbb{R}$: Real
- $\mathbb{C}$: Complex
- $f_Y(y)$: PDF - Probability Density Function
- $F_Y(y)$: CDF - Cumulative Density Function
- $Q(\cdot)$: Q-function

Gaussian or Normal Random Variables

A random variable (RV) X is called a Gaussian or Normal random variable if its probability density function (PDF) is given by:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad x \in \mathbb{R}, \mu \in \mathbb{R}, \text{ and } \sigma > 0 \quad (1)$$

$X \sim \mathcal{N}(\mu, \sigma^2)$: Symbolic notation that X is Gaussian or normal RV with parameters μ, σ^2 .

A RV Y distributed as $\mathcal{N}(0, 1)$ is referred to as a standard normal RV.

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}}, \quad -\infty < y < \infty \quad (2)$$

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{y^2}{2}} dy = 1 \quad (3)$$

$$\mu_Y = \mathbb{E}[Y] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y e^{-\frac{y^2}{2}} dy = 0 \quad (4)$$

$$\begin{aligned} \mathbb{E}[Y^2] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y^2 e^{-\frac{y^2}{2}} dy \\ &= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} y^2 e^{-\frac{y^2}{2}} dy \end{aligned}$$

put $v = \frac{y^2}{2} \Rightarrow dv = y dy$

$$\begin{aligned} \mathbb{E}[Y^2] &= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} \sqrt{2v} e^{-v} dv \\ &= \frac{2}{\sqrt{\pi}} \int_0^{\infty} v^{\frac{1}{2}} e^{-v} dv \end{aligned} \quad (5)$$

Gamma Function

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt, \quad \Re(z) > 0 \quad (6)$$

$$\Gamma(1) = \int_0^\infty e^{-t} dt = 1$$

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-\frac{1}{2}} e^{-t} dt$$

put $t = \frac{v^2}{2} \Rightarrow dt = v dv$

$$\begin{aligned} \therefore \Gamma\left(\frac{1}{2}\right) &= \int_0^\infty \frac{\sqrt{2}}{v} e^{-\frac{v^2}{2}} v dv \\ &= \sqrt{2} \int_0^\infty e^{-\frac{v^2}{2}} dv \\ &= 2\sqrt{\pi} \int_0^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{v^2}{2}} dv \end{aligned} \quad (7)$$

$$\begin{aligned} \Gamma\left(\frac{1}{2}\right) &= \sqrt{\pi} \\ \Gamma(z+1) &= z\Gamma(z) \end{aligned} \quad (8)$$

For positive integer k ,

$$\begin{aligned} \Gamma(k+1) &= k\Gamma(k) \\ &= k(k-1)\Gamma(k-1) \\ &= k(k-1)(k-2) \cdots 2 \cdot 1 \cdot \Gamma(1) \\ &= k! \end{aligned} \quad (9)$$

Using eqns. (7) and (8) in eqn. (5),

$$\begin{aligned} \therefore \mathbb{E}[Y^2] &= \frac{2}{\sqrt{\pi}} \int_0^\infty v^{\frac{1}{2}} e^{-v} dv \\ &= \frac{2}{\sqrt{\pi}} \int_0^\infty v^{\frac{3}{2}-1} e^{-v} dv \\ &= \frac{2}{\sqrt{\pi}} \Gamma\left(\frac{3}{2}\right) = \frac{2}{\sqrt{\pi}} \frac{1}{2} \Gamma\left(\frac{1}{2}\right) \\ &= 1 \end{aligned} \quad (10)$$

CDF of the Standard Normal RV

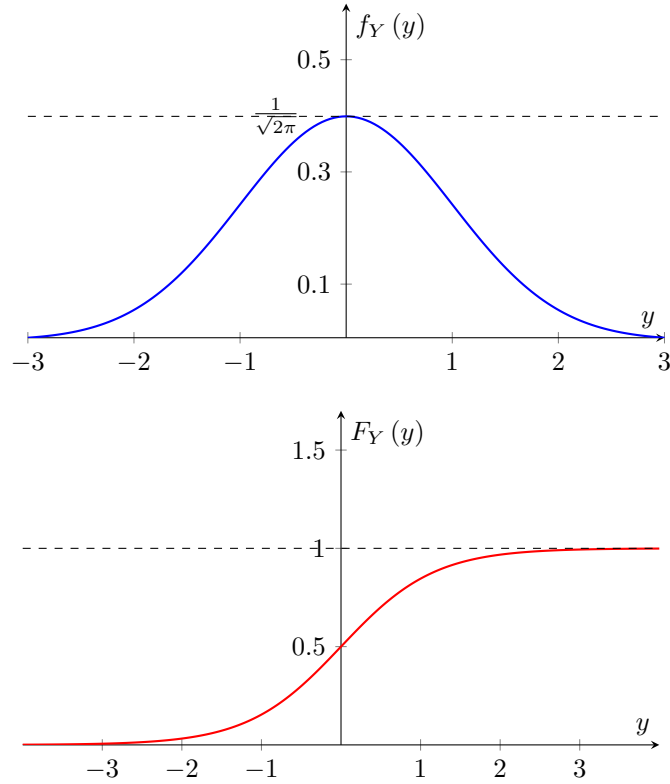
$$\begin{aligned} F_Y(y) &= \Pr(Y \leq y) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^y e^{-\frac{v^2}{2}} dv \\ &= 1 - \frac{1}{\sqrt{2\pi}} \int_y^\infty e^{-\frac{v^2}{2}} dv \\ F_Y(y) &= 1 - Q(y) \end{aligned} \quad (11)$$

Gaussian Q-Function

$$Q(y) \triangleq \frac{1}{\sqrt{2\pi}} \int_y^\infty e^{-\frac{v^2}{2}} dv \quad (12)$$

Properties

- $Q(-y) = 1 - Q(y)$
- $Q(-\infty) = 1$
- $Q(0) = \frac{1}{2}$
- $Q(\infty) = 0$



To find the PDF of $aY + b$ when $Y \sim \mathcal{N}(0, 1)$

Let $V = aY + B$, where a & b are constants $\in \mathbb{R}$.

$$\begin{aligned}
 F_V(v) &= \Pr(V \leq v) \\
 &= \Pr(aY + b \leq v) \\
 &= \Pr(aY \leq v - b) \\
 &= \Pr\left(Y \leq \frac{v - b}{a}\right) \quad \text{if } a > 0 \\
 &= \Pr\left(Y \geq \frac{v - b}{a}\right) \quad \text{if } a < 0
 \end{aligned} \tag{13}$$

If $a > 0$,

$$F_V(v) = F_Y\left(\frac{v - b}{a}\right)$$

Differentiating both sides wrt v , we get,

$$\begin{aligned}
 f_V(v) &= \frac{d\left(\frac{v-b}{a}\right)}{dv} \cdot \frac{F_Y\left(\frac{v-b}{a}\right)}{d\left(\frac{v-b}{a}\right)} \\
 f_V(v) &= \frac{1}{a} f_Y\left(\frac{v - b}{a}\right)
 \end{aligned} \tag{14}$$

If $a < 0$,

$$\begin{aligned} F_V(v) &= \Pr\left(Y \geq \frac{v-b}{a}\right) \\ &= 1 - \Pr\left(Y \leq \frac{v-b}{a}\right) = 1 - F_Y\left(\frac{v-b}{a}\right) \end{aligned}$$

(since there are no impulses in $f_Y(\cdot)$ and hence there are no discontinuities in $F_Y(\cdot)$)
Differentiating both sides wrt v , we get,

$$f_V(v) = -\frac{1}{a}f_Y\left(\frac{v-b}{a}\right) \quad (15)$$

Combining eqns. (14) and (15), we get,

$$f_V(v) = \frac{1}{|a|}f_Y\left(\frac{v-b}{a}\right) \quad (16)$$

$$\Rightarrow f_V(v) = \frac{1}{\sqrt{2\pi}|a|}e^{-\frac{1}{2}\left(\frac{v-b}{|a|}\right)^2}, \quad v \in \mathbb{R} \quad (17)$$

$$\text{or } f_V(v) = \frac{1}{\sqrt{2\pi}a^2}e^{-\frac{(v-b)^2}{2a^2}}, \quad v \in \mathbb{R}$$

Mean μ_V ,

$$\mu_V = \mathbb{E}[aY + b] = a \cdot \mathbb{E}[Y + b] = b \quad (18)$$

Variance $\text{Var}(V)$,

$$\begin{aligned} \text{Var}(V) &= \mathbb{E}\left[(V - \mu_V)^2\right] \\ &= \mathbb{E}\left[(aY + b - b)^2\right] \\ &= a^2\mathbb{E}[Y^2] = a^2 \end{aligned} \quad (19)$$

$$\therefore V \sim \mathcal{N}(b, a^2), \quad Y = \frac{V - b}{a} \sim \mathcal{N}(0, 1)$$

Note

If $Y \sim \mathcal{N}(\mu, \sigma^2)$, then $V = aY + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$.

Moments

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $Y = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$.

$$\mathbb{E}[Y^k] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \underbrace{y^k e^{-\frac{y^2}{2}}}_{\substack{\text{odd function of } y \text{ for } k = 1, 3, 5, \dots \\ \text{even function of } y \text{ for } k = 2, 4, 6, \dots}} dy \quad (20)$$

$$\mathbb{E}[Y^k] = 0 \quad \text{for } k = 1, 3, 5, \dots \quad (21)$$

For $k = 2l$, $l = 1, 2, 3, \dots$

$$\begin{aligned} \mathbb{E}[Y^k] &= \mathbb{E}[Y^{2l}] \\ &= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} y^{2l} e^{-\frac{y^2}{2}} dy \end{aligned}$$

put $v = \frac{y^2}{2} \Rightarrow dv = y dy$,

$$\begin{aligned} \mathbb{E}[Y^{2l}] &= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} (\sqrt{2v})^{2l} e^{-v} y^{-1} dv \\ &= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} (\sqrt{2v})^{2l-1} e^{-v} dv \\ &= \frac{2^{l+\frac{1}{2}}}{\sqrt{2\pi}} \int_0^{\infty} v^{l-\frac{1}{2}} e^{-v} dv \end{aligned}$$

Using eqns. (6) and (8)

$$\mathbb{E} [Y^{2l}] = \frac{2^l}{\sqrt{\pi}} \Gamma \left(l + \frac{1}{2} \right), \quad l = 1, 2, 3, \dots \quad (22)$$

Therefore,

$$\mathbb{E} [(X - \mu)^k] = \begin{cases} 0 & k = 1, 3, 5, \dots \\ \frac{2^{\frac{k}{2}}}{\sqrt{\pi}} \Gamma \left(\frac{k+1}{2} \right) \sigma^k & k = 2, 4, 6, \dots \end{cases} \quad (23)$$

Lecture 2

Types of Random Variables

Scalar $\mathbb{R}$, Vector $\mathbb{R}$, Scalar $\mathbb{C}$ and Vector $\mathbb{C}$

Characteristic Function

It is sometimes easier to characterise a random variable by its CF rather than by its PDF. Also called moment generating function (MGF).

$$\begin{aligned}
 \varphi_X(j\omega) &= \mathbb{E}[e^{j\omega X}] \\
 &= \mathbb{E}\left[\sum_{k=0}^{\infty} \frac{(j\omega X)^k}{k!}\right] \\
 &= \sum_{k=0}^{\infty} \frac{(j\omega)^k}{k!} \mathbb{E}[X^k] \\
 \varphi_X(j\omega) &= 1 + j\omega \mathbb{E}[X] + \frac{(j\omega)^2}{2} \mathbb{E}[X^2] + \dots
 \end{aligned} \tag{24}$$

To find $\mathbb{E}[X]$,

$$\begin{aligned}
 \left. \frac{d\varphi_X(j\omega)}{d\omega} \right|_{\omega=0} &= 0 + j\mathbb{E}[X] + 0 + \dots \\
 \frac{1}{j} \left. \frac{d\varphi_X(j\omega)}{d\omega} \right|_{\omega=0} &= \mathbb{E}[X]
 \end{aligned} \tag{25}$$

Similarly for $\mathbb{E}[X^2]$,

$$\begin{aligned}
 \left. \frac{d^2\varphi_X(j\omega)}{d\omega^2} \right|_{\omega=0} &= 0 + 0 + j^2 \mathbb{E}[X^2] + 0 + \dots \\
 \frac{1}{j^2} \left. \frac{d^2\varphi_X(j\omega)}{d\omega^2} \right|_{\omega=0} &= \mathbb{E}[X^2]
 \end{aligned}$$

In general,

$$\left. \frac{1}{j^k} \frac{d^k\varphi_X(j\omega)}{d\omega^k} \right|_{\omega=0} = \mathbb{E}[X^k] \tag{26}$$

Gaussian Characteristic Function

$$\begin{aligned}
 \varphi_X(j\omega) &= \mathbb{E}[e^{j\omega X}], \quad \text{where } X \sim \mathcal{N}(\mu, \sigma^2) \\
 &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{j\omega x} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx
 \end{aligned} \tag{27}$$

Take

$$\begin{aligned}
 j\omega x - \frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 &= -\frac{1}{2} \left(\frac{x^2 + \mu^2 - 2\mu x}{\sigma^2} - 2j\omega x \right) \\
 &= -\frac{1}{2\sigma^2} (x^2 + \mu^2 - 2\mu x - 2j\omega x \sigma^2) \\
 &= -\frac{1}{2\sigma^2} [x^2 + \mu^2 - 2(\mu + j\omega\sigma^2)x + \mu^2] \\
 &= -\frac{1}{2\sigma^2} [x^2 + \mu^2 - 2(\mu + j\omega\sigma^2)x + (\mu + j\omega\sigma^2)^2 + \mu^2 - (\mu + j\omega\sigma^2)^2]
 \end{aligned}$$

$$\begin{aligned}
&= -\frac{1}{2\sigma^2} [x - (\mu + j\omega\sigma^2)]^2 - \frac{1}{2\sigma^2} [\mu^2 - (\mu + j\omega\sigma^2)] \\
&= -\frac{1}{2\sigma^2} [x - (\mu + j\omega\sigma^2)]^2 - \frac{1}{2\sigma^2} (\mu^2 - \mu^2 - 2j\omega\sigma^2 + \omega^2\sigma^4) \\
&= -\frac{1}{2} \left[\frac{x - (\mu + j\omega\sigma^2)}{\sigma} \right]^2 + \frac{1}{2} (2j\omega\sigma^2 - \omega^2\sigma^2)
\end{aligned}$$

So,

$$\begin{aligned}
\varphi_X(j\omega) &= e^{\frac{1}{2}(2j\omega\mu - \omega^2\sigma^2)} \cdot \underbrace{\frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2} \left[\frac{x - (\mu + j\omega\sigma^2)}{\sigma} \right]^2} dx}_{=1} \\
\therefore \varphi_X(j\omega) &= e^{j\omega\mu - \frac{1}{2}\omega^2\sigma^2}, \quad \text{when } X \sim \mathcal{N}(\mu, \sigma^2)
\end{aligned} \tag{28}$$

For the standard normal RV Y , $\mu = 0$ and $\sigma^2 = 1$,

$$\therefore \varphi_Y(j\omega) = e^{-\frac{1}{2}\omega^2} \tag{29}$$

Note

- $X \sim \mathcal{N}(\mu, \sigma^2)$, $Y \sim \mathcal{N}(0, 1)$

$$\Rightarrow Y = \frac{X - \mu}{\sigma} \Rightarrow X = \sigma Y + \mu$$

$$\begin{aligned}
\varphi_X(j\omega) &= \mathbb{E}[e^{j\omega X}] = \mathbb{E}[e^{j\omega(\sigma Y + \mu)}] \\
&= e^{j\omega\mu} \mathbb{E}[e^{j\omega\sigma Y}] \\
&= e^{j\omega\mu} \varphi_Y(j\omega\sigma) \\
&= e^{j\omega\mu} e^{-\frac{1}{2}\omega^2\sigma^2} = e^{j\omega\mu - \frac{1}{2}\omega^2\sigma^2}
\end{aligned}$$

- If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $\sigma^2 \rightarrow 0$, then

$$\begin{aligned}
\varphi_X(j\omega) &\rightarrow e^{j\omega\mu}, \\
f_X(x) &\rightarrow \delta(x - \mu)
\end{aligned}$$

Jointly Gaussian Random Variables: The Multivariate Gaussian Distribution

Random variables $X_1, X_2, \dots, X_L$ are called jointly Gaussian or jointly normal random variables if their joint PDF is given by,

$$\begin{aligned}
f_{X_1, X_2, \dots, X_L}(x_1, x_2, \dots, x_L) &= \frac{1}{(2\pi)^{\frac{L}{2}} |\mathbf{K}|^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2} \sum_{k=1}^L \sum_{l=1}^L (x_k - \mu_k) (\mathbf{K}^{-1})_{kl} (x_l - \mu_l) \right\}, \\
&\quad -\infty < x_1, x_2, \dots, x_L < \infty \text{ and } -\infty < \mu_1, \mu_2, \dots, \mu_L < \infty \quad (30)
\end{aligned}$$

- $\mathbf{K}$ is a real-valued $L \times L$ positive definite matrix.
- $(\mathbf{K}^{-1})_{kl}$ denotes the element in the k -th row and l -th column of $\mathbf{K}^{-1}$.

Denoting the random vector $\mathbf{X}$ as,

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_L \end{bmatrix}$$

and the vectros $\mathbf{x}$ and $\boldsymbol{\mu}$ as,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_L \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_L \end{bmatrix}$$

We can express the joint PDF of $X_1, X_2, \dots, X_L$ as the PDF of the random vector $\mathbf{X}$, which is given by,

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{L}{2}} |\mathbf{K}|^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^{\top} \mathbf{K}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$

$\mathbf{x} \in \mathbb{R}^L, \boldsymbol{\mu} \in \mathbb{R}^L, \mathbf{K} \in \mathbb{R}^{L \times L}$ and $\mathbf{K}$ is positive definite. (31)

Note that $\mathbf{K}$ is real-valued and positive definite implies,

- $\det(\mathbf{K}) = |\mathbf{K}| > 0$
- $\mathbf{K}$ is symmetric ($\mathbf{K} = \mathbf{K}^{\top}$)
- All eigenvalues of $\mathbf{K}$ are real and positive.

A matrix $\mathbf{M} \in \mathbb{R}^{L \times L}$ is positive definite when for all $\mathbf{x} \in \mathbb{R}^L - \{\mathbf{0}_{L \times 1}\}$, $\mathbf{x}^{\top} \mathbf{M} \mathbf{x} > 0$.

$$\mathbb{E}[X_k] = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} x_k f_{X_1, \dots, X_L}(x_1, \dots, x_L) dx_1 \dots dx_L = \mu_k, \quad k = 1, 2, \dots, L \quad (32)$$

$$\begin{aligned} \mathbb{E}[(X_k - \mu_k)(X_l - \mu_l)] &= \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} (x_k - \mu_k)(x_l - \mu_l) f_{X_1, \dots, X_L}(x_1, \dots, x_L) dx_1 \dots dx_L \\ &= (\mathbf{K}^{-1})_{kl}, \end{aligned}$$

$k = 1, \dots, L$ and $l = 1, \dots, L$ (33)

$$\therefore \mathbb{E}[\mathbf{X}] = \boldsymbol{\mu} : \text{Mean vector} \quad (34)$$

$$\mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^{\top}] = \mathbf{K} : \text{Covariance matrix} \quad (35)$$

$$\mathbb{E}[\mathbf{X}\mathbf{X}^{\top}] = \mathbf{K} + \boldsymbol{\mu}\boldsymbol{\mu}^{\top} = \mathbf{R} : \text{Correlation matrix} \quad (36)$$

We say that $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{K})$

$\mathbf{X}$ is a Gaussian random vector with mean vector $\boldsymbol{\mu}$ and covariance matrix $\mathbf{K}$.

Characteristic function of $\mathbf{X}$ or joint characteristic function of $X_1, X_2, \dots, X_L$

$$\begin{aligned} \varphi_{X_1, \dots, X_L}(j\omega_1, \dots, j\omega_L) &= \mathbb{E}[e^{j\omega_1 X_1 + \dots + j\omega_L X_L}] \\ \varphi_{\mathbf{X}}(j\boldsymbol{\omega}) &= \mathbb{E}[e^{j\boldsymbol{\omega}^{\top} \mathbf{X}}] \\ \varphi_{\mathbf{X}}(j\boldsymbol{\omega}) &= e^{j\boldsymbol{\omega}^{\top} \boldsymbol{\mu} - \frac{1}{2} \boldsymbol{\omega}^{\top} \mathbf{K} \boldsymbol{\omega}} \end{aligned} \quad (37)$$

We can also write as,

$$\varphi_{X_1, \dots, X_L}(j\omega_1, \dots, j\omega_L) = e^{j \sum_{k=1}^L \omega_k \mu_k - \frac{1}{2} \sum_{k=1}^L \sum_{l=1}^L \omega_k (\mathbf{K})_{kl} \omega_l} \quad (38)$$

when $L = 1$ we have $\boldsymbol{\mu} = \mu$ and $\mathbf{K} = \sigma^2$

$$\Rightarrow \varphi_X(j\omega) = e^{j\omega\mu - \frac{1}{2}\omega^2\sigma^2} \quad (39)$$

From the joint characteristic function, we can easily obtain the marginal distributions of X_k , $k = 1, \dots, L$ putting $\omega_1 = \dots = \omega_{k-1} = \omega_{k+1} = \dots = \omega_L = 0$ in $\varphi_{\mathbf{X}}(j\boldsymbol{\omega})$

$$\begin{aligned} \varphi_{X_k}(j\omega_k) &= e^{j\omega_k \mu_k - \frac{1}{2} \omega_k^2 (\mathbf{K})_{kk}} \\ \therefore X_k &\sim \mathcal{N}(\mu_k, (\mathbf{K})_{kk}), \quad k = 1, \dots, L \end{aligned} \quad (40)$$

If $X_1, \dots, X_L$ are jointly Gaussian then X_k is marginally Gaussian for $k = 1, \dots, L$. However, the reverse is not true!

Lecture 3

Whitening

Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{K})$ be an $L \times 1$ Gaussian vector and $\mathbf{V} = \mathbf{A}\mathbf{X} + \mathbf{b}$, where $\mathbf{A} \in \mathbb{R}^{L \times L}$, $\mathbf{b} \in \mathbb{R}^L$ and $\det(\mathbf{A}) \neq 0$. Then

$$\mathbb{E}[\mathbf{V}] = \mathbb{E}[\mathbf{A}\mathbf{X} + \mathbf{b}] = \mathbf{A}\boldsymbol{\mu} + \mathbf{b} \quad (41)$$

$$\begin{aligned} \text{Var}(\mathbf{V}) &= \mathbb{E}[(\mathbf{V} - \boldsymbol{\mu}_{\mathbf{V}})(\mathbf{V} - \boldsymbol{\mu}_{\mathbf{V}})^\top] \\ &= \mathbb{E}[(\mathbf{A}\mathbf{X} + \mathbf{b} - \mathbf{A}\boldsymbol{\mu} - \mathbf{b})(\mathbf{A}\mathbf{X} + \mathbf{b} - \mathbf{A}\boldsymbol{\mu} - \mathbf{b})^\top] \\ &= \mathbb{E}[\mathbf{A}(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top \mathbf{A}^\top] \\ &= \mathbf{A}\mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top] \mathbf{A}^\top \\ &= \mathbf{A}\mathbf{K}\mathbf{A}^\top \end{aligned} \quad (42)$$

$$\therefore \mathbf{V} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\mathbf{K}\mathbf{A}^\top)$$

Therefore, if $\mathbf{b} = -\mathbf{A}\boldsymbol{\mu}$ and $\mathbf{K} = (\mathbf{A}^\top \mathbf{A})^{-1}$, then $\mathbf{V} = \mathbf{A}\mathbf{X} + \mathbf{b}$, where $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{K})$ has a $\mathcal{N}(\mathbf{0}, \mathbf{I}_L)$ distribution.

When $\mathbf{V} = \begin{bmatrix} V_1 \\ \vdots \\ V_L \end{bmatrix}$ has a $\mathcal{N}(\mathbf{0}, \mathbf{I}_L)$ distribution we can say that $V_1, \dots, V_L$ are iid $\mathcal{N}(0, 1)$ random variables.

A $\mathcal{N}(\boldsymbol{\mu}, \alpha \mathbf{I})$ Gaussian random vector, where $\alpha > 0$, is called a white random vector.

If $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{K})$ is an $L \times 1$ Gaussian random vector such that $\mathbf{K} = \mathbf{B}\mathbf{B}^\top$, where $\mathbf{B}$ is an $L \times L$ real-valued non singular matrix, then $\mathbf{B}^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_L)$ and $\mathbf{B}^{-1}\mathbf{X} \sim \mathcal{N}(\mathbf{B}^{-1}\boldsymbol{\mu}, \mathbf{I}_L)$.

Transforming a Gaussian random vector with correlated elements to another Gaussian random vector with covariance matrix equation to the identity matrix or its scaled version by a linear transformation is called whitening.

Whitening by Eigendecomposition

Let $\lambda_1, \lambda_2, \dots, \lambda_L$ be the eigenvalues of $\mathbf{K}$ (all of them may not be distinct). Since $\mathbf{K}$ is a real-valued symmetric positive definite matrix, $\lambda_1, \dots, \lambda_L$ are real and positive, and it is possible to find a set $\{\mathbf{e}_1, \dots, \mathbf{e}_L\}$ of L linearly independent eigenvectors. $\mathbf{e}_i$ is an eigenvector corresponding to eigenvalue λ_i , $i = 1, 2, \dots, L$.

Thus,

$$\mathbf{K}\mathbf{e}_i = \lambda_i \mathbf{e}_i, \quad i = 1, \dots, L \quad (43)$$

$$\mathbf{K}[\mathbf{e}_i \quad \dots \quad \mathbf{e}_L] = [\mathbf{e}_i \quad \dots \quad \mathbf{e}_L] \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_L \end{bmatrix} \quad (44)$$

$$\text{Let } \mathbf{E} = [\mathbf{e}_1 \quad \dots \quad \mathbf{e}_L] \text{ and } \boldsymbol{\Lambda} = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_L \end{bmatrix} = \text{diag}(\lambda_1, \dots, \lambda_L), \text{ we get} \quad (45)$$

$$\mathbf{K}\mathbf{E} = \mathbf{E}\boldsymbol{\Lambda}$$

Since the columns of $\mathbf{E}$ are linearly independent, $\mathbf{E}$ is full-rank (rank L), and is therefore invertible (non singular).

$$\therefore \mathbf{K} = \mathbf{E}\boldsymbol{\Lambda}\mathbf{E}^{-1} \quad (46)$$

Since $\mathbf{K}$ is a symmetric matrix, the linearly independent eigenvectors are also orthogonal.

Note that a real-valued symmetric matrix always has real eigenvalues, and hence real-valued eigenvectors.

In addition, we can normalize the eigenvectors so that $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_L\}$ is an orthonormal set.

$$\Rightarrow \mathbf{e}_i^\top \mathbf{e}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (47)$$

$$\begin{bmatrix} \mathbf{e}_1^\top \\ \vdots \\ \mathbf{e}_L^\top \end{bmatrix} \begin{bmatrix} \mathbf{e}_1 & \dots & \mathbf{e}_L \end{bmatrix} = \mathbf{I}_L \quad (48)$$

i.e. $\mathbf{E}^\top \mathbf{E} = \mathbf{I}_L$

Since $\mathbf{E}$ is non-singular, this also implies $\mathbf{E}^\top = \mathbf{E}^{-1}$

$$\Rightarrow \mathbf{E}\mathbf{E}^\top = \sum_{i=1}^L \mathbf{e}_i \mathbf{e}_i^\top = \mathbf{I}_L \quad (49)$$

$$\therefore \mathbf{K} = \mathbf{E}\mathbf{\Lambda}\mathbf{E}^{-1} = \mathbf{E}\mathbf{\Lambda}\mathbf{E}^\top \quad (50)$$

Note that $\mathbf{K} = \mathbf{K}^\top$ implies

$$\begin{aligned} \mathbf{E}\mathbf{\Lambda}\mathbf{E}^{-1} &= (\mathbf{E}\mathbf{\Lambda}\mathbf{E}^{-1})^\top \\ &= (\mathbf{E}^{-1})^\top \mathbf{\Lambda} \mathbf{E}^\top \end{aligned}$$

which is consistent with the fact that $\mathbf{E}^\top = \mathbf{E}^{-1}$.

Denoting

$$\mathbf{\Lambda}^{\frac{1}{2}} = \text{diag}(\lambda_1^{\frac{1}{2}}, \dots, \lambda_L^{\frac{1}{2}}) \quad (51)$$

We can write

$$\begin{aligned} \mathbf{K} &= \mathbf{E}\mathbf{\Lambda}\mathbf{E}^\top \\ &= \mathbf{E}\mathbf{\Lambda}^{\frac{1}{2}}\mathbf{\Lambda}^{\frac{1}{2}}\mathbf{E}^\top \\ &= (\mathbf{E}\mathbf{\Lambda}^{\frac{1}{2}})(\mathbf{E}\mathbf{\Lambda}^{\frac{1}{2}})^\top \end{aligned} \quad (52)$$

Since $\mathbf{K} = \mathbf{B}\mathbf{B}^\top$

$$\therefore \mathbf{B} = \mathbf{E}\mathbf{\Lambda}^{\frac{1}{2}} \quad (53)$$

and

$$\mathbf{B}^{-1} = \mathbf{\Lambda}^{-\frac{1}{2}}\mathbf{E}^{-1} = \mathbf{\Lambda}^{-\frac{1}{2}}\mathbf{E}^\top \quad (54)$$

Example

$$\mathbf{K} = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

Eignevalues :

$$\begin{aligned} &\begin{vmatrix} 3-\lambda & 1 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 1 & 3-\lambda \end{vmatrix} = 0 \\ (3-\lambda) \left[(3-\lambda)^2 - 1 \right] - 1 \left[(3-\lambda) - 1 \right] + 1 \left[1 - (3-\lambda) \right] &= 0 \\ (\lambda-2)(\lambda-2)(\lambda-5) &= 0 \end{aligned}$$

$$\lambda_1 = 2, \lambda_2 = 2, \lambda_3 = 5$$

Eigenvectors :

Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be an eigenvector for eigenvalue 2.

$$\mathbf{K}\mathbf{x} = 2\mathbf{x} \Rightarrow 3x_1 + x_2 + x_3 = 2x_1 \quad (\text{A})$$

$$x_1 + 3x_2 + x_3 = 2x_2 \quad (\text{B})$$

$$x_1 + x_2 + 3x_3 = 2x_3 \quad (\text{C})$$

From eqns. (A), (B) and (C) $\Rightarrow x_1 + x_2 + x_3 = 0$

Choose $\mathbf{e}_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$, $\mathbf{e}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$. $\{\mathbf{e}_1, \mathbf{e}_2\}$ form an orthonormal set. The choice of $\mathbf{e}_1, \mathbf{e}_2$ is not unique.

Let $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ be an eigenvector for eigenvalue 5.

$$\mathbf{K}\mathbf{y} = 2\mathbf{y} \Rightarrow 3y_1 + y_2 + y_3 = 5y_1 \quad (\text{D})$$

$$y_1 + 3y_2 + y_3 = 5y_2 \quad (\text{E})$$

$$y_1 + y_2 + 3y_3 = 5y_3 \quad (\text{F})$$

From eqns. (D), (E) and (F) $\Rightarrow y_1 = y_2 = y_3$

Choose $\mathbf{e}_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

$$\therefore \mathbf{E} = [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3]$$

$$\mathbf{E} = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ -\frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \end{bmatrix}$$

$$\mathbf{E}^{-1} = \mathbf{E}^\top = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

$$\mathbf{\Lambda}^{-\frac{1}{2}} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & \frac{1}{\sqrt{5}} \end{bmatrix}$$

$$\mathbf{B} = \mathbf{E}\mathbf{\Lambda}^{\frac{1}{2}}$$

$$\mathbf{B}^{-1} = \mathbf{\Lambda}^{-\frac{1}{2}}\mathbf{E}^\top = \begin{bmatrix} \frac{1}{\sqrt{12}} & \frac{1}{\sqrt{12}} & -\frac{2}{\sqrt{12}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{15}} & \frac{1}{\sqrt{15}} & \frac{1}{\sqrt{15}} \end{bmatrix}$$

Therefore, if $\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix} \right)$

$$\mathbf{Y} = \mathbf{B}^{-1}\mathbf{X} = \begin{bmatrix} \frac{1}{\sqrt{12}} & \frac{1}{\sqrt{12}} & -\frac{2}{\sqrt{12}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{15}} & \frac{1}{\sqrt{15}} & \frac{1}{\sqrt{15}} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$$

$$\mathbf{Y} = \begin{bmatrix} \frac{X_1 + X_2}{\sqrt{12}} - \frac{2X_3}{\sqrt{12}} \\ \frac{X_1 - X_2}{2} \\ \frac{X_1 + X_2 + X_3}{\sqrt{15}} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right)$$

Lecture 4

The Complex Gaussian Random Vector

Let $\mathbf{X}$ and $\mathbf{Y}$ be $L \times 1$ real random vectors, which are jointly Gaussian.

Let $\mathbb{E}[\mathbf{X}] = \boldsymbol{\mu}_X$ and $\mathbb{E}[\mathbf{Y}] = \boldsymbol{\mu}_Y$.

$$\underbrace{\mathbf{K}}_{\text{Covariance matrix}} = \mathbb{E} \left[\begin{bmatrix} \mathbf{X} - \boldsymbol{\mu}_X \\ \mathbf{Y} - \boldsymbol{\mu}_Y \end{bmatrix} \begin{bmatrix} (\mathbf{X} - \boldsymbol{\mu}_X)^\top & (\mathbf{Y} - \boldsymbol{\mu}_Y)^\top \end{bmatrix} \right] = \begin{bmatrix} \mathbf{K}_{XX} & \mathbf{K}_{XY} \\ \mathbf{K}_{YX} & \mathbf{K}_{YY} \end{bmatrix} \quad (55)$$

Thus,

$$\left. \begin{aligned} \mathbf{K}_{XX} &= \mathbb{E} \left[(\mathbf{X} - \boldsymbol{\mu}_X) (\mathbf{X} - \boldsymbol{\mu}_X)^\top \right] \\ \mathbf{K}_{YY} &= \mathbb{E} \left[(\mathbf{Y} - \boldsymbol{\mu}_Y) (\mathbf{Y} - \boldsymbol{\mu}_Y)^\top \right] \end{aligned} \right\} \begin{array}{l} \text{Autocovariance or Covariance} \\ \text{matrix of } \mathbf{X} \text{ or } \mathbf{Y} \end{array} \quad (56)$$

$$\left. \begin{aligned} \mathbf{K}_{XY} &= \mathbb{E} \left[(\mathbf{X} - \boldsymbol{\mu}_X) (\mathbf{Y} - \boldsymbol{\mu}_Y)^\top \right] \\ \mathbf{K}_{YX} = \mathbf{K}_{XY}^\top &= \mathbb{E} \left[(\mathbf{Y} - \boldsymbol{\mu}_Y) (\mathbf{X} - \boldsymbol{\mu}_X)^\top \right] \end{aligned} \right\} \begin{array}{l} \text{Covariance-covariance matrix} \\ \text{of } \mathbf{X} \text{ and } \mathbf{Y} \end{array} \quad (57)$$

Note

- $\mathbf{K}$, $\mathbf{K}_{XX}$ and $\mathbf{K}_{YY}$ are covariance matrices, hence they are positive definite.
- $\mathbf{K}_{XY}$ is a cross covariance matrix, and in general, is not positive definite.

$$\begin{aligned} \boldsymbol{\mu}_X, \boldsymbol{\mu}_Y &\in \mathbb{R}^L \\ \mathbf{K}_{XX}, \mathbf{K}_{YY}, \mathbf{K}_{XY} &\in \mathbb{R}^{L \times L} \\ \mathbf{K} &\in \mathbb{R}^{2L \times 2L} \end{aligned}$$

Joint PDF of $\mathbf{X}$ and $\mathbf{Y}$

$$f_{\mathbf{X}, \mathbf{Y}}(\mathbf{x}, \mathbf{y}) = \frac{1}{(2\pi)^L |\mathbf{K}|^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2} \begin{bmatrix} (\mathbf{x} - \boldsymbol{\mu}_X)^\top & (\mathbf{y} - \boldsymbol{\mu}_Y)^\top \end{bmatrix} \mathbf{K}^{-1} \begin{bmatrix} \mathbf{x} - \boldsymbol{\mu}_X \\ \mathbf{y} - \boldsymbol{\mu}_Y \end{bmatrix} \right\} \quad \mathbf{x}, \mathbf{y} \in \mathbb{R}^L \quad (58)$$

Let $\mathbf{Z} = \mathbf{X} + j\mathbf{Y}$ denote the $L \times 1$ complex Gaussian random vector.

Thus,

$$\begin{aligned} \mathbf{X} &= \Re(\mathbf{Z}) = \frac{1}{2} (\mathbf{Z} + \bar{\mathbf{Z}}) \\ \mathbf{Y} &= \Im(\mathbf{Z}) = \frac{1}{2j} (\mathbf{Z} - \bar{\mathbf{Z}}) \end{aligned} \quad (59)$$

$$\mathbb{E}[\mathbf{Z}] = \boldsymbol{\mu}_Z = \boldsymbol{\mu}_X + j\boldsymbol{\mu}_Y \quad (60)$$

Let $\mathbf{Z}' = \mathbf{Z} - \boldsymbol{\mu}_Z$, $\mathbf{X}' = \mathbf{X} - \boldsymbol{\mu}_X$ and $\mathbf{Y}' = \mathbf{Y} - \boldsymbol{\mu}_Y$

$$\left. \begin{aligned} \mathbf{K}_Z &= \mathbb{E} \left[(\mathbf{Z} - \boldsymbol{\mu}_Z) (\mathbf{Z} - \boldsymbol{\mu}_Z)^\mathbf{H} \right] \\ &= \mathbb{E} \left[\mathbf{Z}' \mathbf{Z}'^\mathbf{H} \right] \\ &= \mathbb{E} \left[(\mathbf{X}' + j\mathbf{Y}') (\mathbf{X}' - j\mathbf{Y}')^\top \right] \\ &= \mathbb{E} \left[\mathbf{X}' \mathbf{X}'^\top \right] + \mathbb{E} \left[\mathbf{Y}' \mathbf{Y}'^\top \right] + j \{ \mathbb{E} \left[\mathbf{Y}' \mathbf{X}'^\top \right] - \mathbb{E} \left[\mathbf{X}' \mathbf{Y}'^\top \right] \} \\ &= \mathbf{K}_{XX} + \mathbf{K}_{YY} + j (\mathbf{K}_{XY}^\top - \mathbf{K}_{XY}) \end{aligned} \right\} \begin{array}{l} \text{Autocovariance or} \\ \text{Covariance matrix of } \mathbf{Z} \end{array} \quad (61)$$

$$\begin{aligned}
\tilde{\mathbf{K}}_{\mathbf{Z}} &= \mathbb{E} \left[(\mathbf{Z} - \boldsymbol{\mu}_{\mathbf{Z}}) (\mathbf{Z} - \boldsymbol{\mu}_{\mathbf{Z}})^{\top} \right] \\
&= \mathbb{E} [\mathbf{Z}' \mathbf{Z}^{\top}] \\
&= \mathbb{E} [(\mathbf{X}' + j\mathbf{Y}') (\mathbf{X}' + j\mathbf{Y}')^{\top}] \\
&= \mathbb{E} [\mathbf{X}' \mathbf{X}'^{\top}] - \mathbb{E} [\mathbf{Y}' \mathbf{Y}'^{\top}] + j \{ \mathbb{E} [\mathbf{Y}' \mathbf{X}'^{\top}] + \mathbb{E} [\mathbf{X}' \mathbf{Y}'^{\top}] \} \\
&= \mathbf{K}_{\mathbf{X}\mathbf{X}} - \mathbf{K}_{\mathbf{Y}\mathbf{Y}} + j (\mathbf{K}_{\mathbf{X}\mathbf{Y}}^{\top} + \mathbf{K}_{\mathbf{X}\mathbf{Y}})
\end{aligned}
\left. \vphantom{\begin{aligned} \tilde{\mathbf{K}}_{\mathbf{Z}} &= \mathbb{E} \left[(\mathbf{Z} - \boldsymbol{\mu}_{\mathbf{Z}}) (\mathbf{Z} - \boldsymbol{\mu}_{\mathbf{Z}})^{\top} \right] \\ &= \mathbb{E} [\mathbf{Z}' \mathbf{Z}^{\top}] \\ &= \mathbb{E} [(\mathbf{X}' + j\mathbf{Y}') (\mathbf{X}' + j\mathbf{Y}')^{\top}] \\ &= \mathbb{E} [\mathbf{X}' \mathbf{X}'^{\top}] - \mathbb{E} [\mathbf{Y}' \mathbf{Y}'^{\top}] + j \{ \mathbb{E} [\mathbf{Y}' \mathbf{X}'^{\top}] + \mathbb{E} [\mathbf{X}' \mathbf{Y}'^{\top}] \} \\ &= \mathbf{K}_{\mathbf{X}\mathbf{X}} - \mathbf{K}_{\mathbf{Y}\mathbf{Y}} + j (\mathbf{K}_{\mathbf{X}\mathbf{Y}}^{\top} + \mathbf{K}_{\mathbf{X}\mathbf{Y}}) \right\} \begin{array}{l} \text{Pseudo-autocovariance or} \\ \text{Pseudo-covariance matrix of } \mathbf{Z} \end{array}
\end{aligned} \tag{62}$$

Thus,

$$\begin{aligned}
\mathbf{K}_{\mathbf{Z}} &= \mathbf{K}_{\mathbf{X}\mathbf{X}} + \mathbf{K}_{\mathbf{Y}\mathbf{Y}} + j (\mathbf{K}_{\mathbf{X}\mathbf{Y}}^{\top} - \mathbf{K}_{\mathbf{X}\mathbf{Y}}) \\
\tilde{\mathbf{K}}_{\mathbf{Z}} &= \mathbf{K}_{\mathbf{X}\mathbf{X}} - \mathbf{K}_{\mathbf{Y}\mathbf{Y}} + j (\mathbf{K}_{\mathbf{X}\mathbf{Y}}^{\top} + \mathbf{K}_{\mathbf{X}\mathbf{Y}})
\end{aligned}$$

$$\Rightarrow \mathbf{K}_{\mathbf{X}\mathbf{X}} = \frac{1}{2} \left\{ \Re(\mathbf{K}_{\mathbf{Z}}) + \Re(\tilde{\mathbf{K}}_{\mathbf{Z}}) \right\} \tag{63}$$

$$\mathbf{K}_{\mathbf{Y}\mathbf{Y}} = \frac{1}{2} \left\{ \Re(\mathbf{K}_{\mathbf{Z}}) - \Re(\tilde{\mathbf{K}}_{\mathbf{Z}}) \right\} \tag{64}$$

$$\mathbf{K}_{\mathbf{X}\mathbf{Y}} = \frac{1}{2} \left\{ \Im(\tilde{\mathbf{K}}_{\mathbf{Z}}) - \Im(\mathbf{K}_{\mathbf{Z}}) \right\} \tag{65}$$

Note that $\mathbf{K}_{\mathbf{Z}}$ is a Hermitian positive definite matrix.

$\mathbf{K}_{\mathbf{Z}}$ is complex-valued and positive definite implies,

- $\det(\mathbf{K}_{\mathbf{Z}}) = |\mathbf{K}_{\mathbf{Z}}| > 0$
- $\mathbf{K}_{\mathbf{Z}}$ is Hermitian (conjugate symmetric) i.e., $\mathbf{K}_{\mathbf{Z}} = \mathbf{K}_{\mathbf{Z}}^{\mathbf{H}} = (\overline{\mathbf{K}_{\mathbf{Z}}})^{\top}$
- All eigenvalues of $\mathbf{K}_{\mathbf{Z}}$ are real and positive.

A matrix $\mathbf{M} \in \mathbb{C}^{L \times L}$ is positive definite when for all $\mathbf{z} \in \mathbb{C}^L - \{\mathbf{0}_{L \times 1}\}$, $\mathbf{z}^{\mathbf{H}} \mathbf{M} \mathbf{z} > 0$.

$\tilde{\mathbf{K}}_{\mathbf{Z}}$ is a symmetric matrix, in general, it is not positive definite.

If in the joint PDF $f_{\mathbf{X}, \mathbf{Y}}(\mathbf{x}, \mathbf{y})$, we substitute

$$\begin{aligned}
\mathbf{X} &= \Re(\mathbf{Z}), \quad \mathbf{Y} = \Im(\mathbf{Z}) \\
\boldsymbol{\mu}_{\mathbf{X}} &= \Re(\boldsymbol{\mu}_{\mathbf{Z}}), \quad \boldsymbol{\mu}_{\mathbf{Y}} = \Im(\boldsymbol{\mu}_{\mathbf{Z}}) \\
\mathbf{K}_{\mathbf{X}\mathbf{X}} &= \frac{1}{2} \Re(\mathbf{K}_{\mathbf{Z}} + \tilde{\mathbf{K}}_{\mathbf{Z}}) \\
\mathbf{K}_{\mathbf{X}\mathbf{Y}} &= \frac{1}{2} \Im(\tilde{\mathbf{K}}_{\mathbf{Z}} - \mathbf{K}_{\mathbf{Z}})
\end{aligned}$$

then we can write

$$f_{\mathbf{X}, \mathbf{Y}}(\mathbf{x}, \mathbf{y}) = f_{\Re(\mathbf{Z}), \Im(\mathbf{Z})}(\Re(\mathbf{z}), \Im(\mathbf{z})) = f_{\mathbf{Z}}(\mathbf{z}) \tag{66}$$

Thus $f_{\mathbf{Z}}(\mathbf{z})$ is the PDF of the complex Gaussian random vector $\mathbf{Z}$.

Therefore, the PDF of a complex Gaussian random vector $\mathbf{Z}$ is the joint PDF of its real and imaginary parts $\mathbf{X}$ and $\mathbf{Y}$, respectively, $\mathbf{X}$ and $\mathbf{Y}$ being jointly Gaussian.

Joint CF of $\mathbf{X}$ and $\mathbf{Y}$

$$\begin{aligned}
\varphi_{\mathbf{X}, \mathbf{Y}}(j\omega_1, j\omega_2) &= \mathbb{E} \left[e^{j(\omega_1^{\top} \mathbf{X} + \omega_2^{\top} \mathbf{Y})} \right] \\
&= \mathbb{E} \left[e^{j \begin{bmatrix} \omega_1^{\top} & \omega_2^{\top} \end{bmatrix} \begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \end{bmatrix}} \right] \\
&= e^{j(\omega_1^{\top} \boldsymbol{\mu}_{\mathbf{X}} + \omega_2^{\top} \boldsymbol{\mu}_{\mathbf{Y}}) - \frac{1}{2}(\omega_1^{\top} \mathbf{K}_{\mathbf{X}\mathbf{X}} \omega_1 + \omega_2^{\top} \mathbf{K}_{\mathbf{Y}\mathbf{Y}} \omega_2 + 2\omega_1^{\top} \mathbf{K}_{\mathbf{X}\mathbf{Y}} \omega_2)}
\end{aligned} \tag{67}$$

Lecture 5

A complex random vector whose pseudo-covariance matrix is zero is called a circular random vector. (A circular random vector need not be Gaussian.)

If the complex Gaussian vector $\mathbf{Z} = \mathbf{X} + j\mathbf{Y}$ is circular then,

$$\tilde{\mathbf{K}}_{\mathbf{Z}} = \mathbf{0}_{L \times L} \quad (68)$$

implying

$$\mathbf{K}_{\mathbf{X}\mathbf{X}} = \mathbf{K}_{\mathbf{Y}\mathbf{Y}}, \quad \mathbf{K}_{\mathbf{X}\mathbf{Y}} = -\mathbf{K}_{\mathbf{Y}\mathbf{X}}^{\top} \quad (69)$$

i.e., $\mathbf{X}$ and $\mathbf{Y}$ have equal covariance matrices, and the cross-covariance matrix of $\mathbf{X}$ and $\mathbf{Y}$ is skew-symmetric. (A real-valued skew-symmetric matrix always has zeros on its diagonals.)

Thus, if $\mathbf{X}$ and $\mathbf{Y}$ are independent real Gaussian vectors with equal covariance matrices, then $\mathbf{K}_{\mathbf{X}\mathbf{Y}} = -\mathbf{K}_{\mathbf{Y}\mathbf{X}}^{\top}$ is also satisfied, and $\mathbf{Z} = \mathbf{X} + j\mathbf{Y}$ is circular.

Note

- In wireless communication, we are interested in complex random vectors that have the circular symmetry property: $\mathbf{Z}$ is circular symmetric if $e^{j\theta}\mathbf{Z}$ has the same distribution of $\mathbf{Z}$ for any θ .
- If Z is complex circular with zero mean, then $e^{j\theta}Z$ will have the same second moments as Z . This is where the term circular comes from: a rotation of this random variable in the complex plane does not change its second moment description.
- When $L = 1$, we have $Z = X + jY$
with $\mathbf{K}_{XX} = \text{Var}(X) = \sigma_X^2$
 $\mathbf{K}_{YY} = \text{Var}(Y) = \sigma_Y^2$
 $\mathbf{K}_{XY} = \text{Cov}(X, Y) = \rho_{XY}\sigma_X\sigma_Y$
If $Z = X + jY$ is a circular complex Gaussian random variable then $\sigma_X^2 = \sigma_Y^2$ and $\rho_{XY} = 0$ implying that X and Y are uncorrelated, and therefore independent.
- For any $L \times 1$ complex random vector (not necessarily Gaussian), with the real and imaginary parts having equal covariance matrices,
uncorrelated real and imaginary parts $\Rightarrow$ circularity for all $L \geq 1$.
but, circularity $\Rightarrow$ uncorrelated real and imaginary parts for $L = 1$ only.

PDF:

$$\begin{aligned} f_{\mathbf{X}, \mathbf{Y}}(\mathbf{x}, \mathbf{y}) &= \frac{1}{(2\pi)^L |\mathbf{K}|^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2} \begin{bmatrix} (\mathbf{x} - \boldsymbol{\mu}_{\mathbf{X}})^{\top} & (\mathbf{y} - \boldsymbol{\mu}_{\mathbf{Y}})^{\top} \end{bmatrix} \mathbf{K}^{-1} \begin{bmatrix} \mathbf{x} - \boldsymbol{\mu}_{\mathbf{X}} \\ \mathbf{y} - \boldsymbol{\mu}_{\mathbf{Y}} \end{bmatrix} \right\} \\ &= \frac{1}{\pi^L |\mathbf{K}_{\mathbf{Z}}|} \exp \left\{ -(\mathbf{z} - \boldsymbol{\mu}_{\mathbf{Z}})^{\text{H}} \mathbf{K}^{-1} (\mathbf{z} - \boldsymbol{\mu}_{\mathbf{Z}}) \right\} \\ &= f_{\mathbf{Z}}(\mathbf{z}) \end{aligned} \quad (70)$$

Thus when $\mathbf{Z}$ is a circular complex Gaussian random vector, we have

$$\begin{aligned} f_{\mathbf{Z}}(\mathbf{z}) &= \frac{1}{\pi^L |\mathbf{K}_{\mathbf{Z}}|} \exp \left\{ -(\mathbf{z} - \boldsymbol{\mu}_{\mathbf{Z}})^{\text{H}} \mathbf{K}^{-1} (\mathbf{z} - \boldsymbol{\mu}_{\mathbf{Z}}) \right\} \quad \mathbf{z} \in \mathbb{C}^L \text{ or } \Re(\mathbf{z}), \Im(\mathbf{z}) \in \mathbb{R}^L \\ \mathbf{Z} &\sim \mathcal{CN}(\boldsymbol{\mu}_{\mathbf{Z}}, \mathbf{K}_{\mathbf{Z}}) \end{aligned} \quad (71)$$

i.e., $\mathbf{Z}$ is a circular complex Gaussian random vector with mean vector $\boldsymbol{\mu}_{\mathbf{Z}}$ and covariance matrix $\mathbf{K}_{\mathbf{Z}}$.

Rician and Rayleigh Random Variables

Let $X_1 \sim \mathcal{N}(\mu_1, \sigma^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma^2)$, X_1 and X_2 are independent. Let $Z = X_1 + jX_2 = Re^{j\Theta}$.

$$\left. \begin{aligned} R &= \sqrt{X_1^2 + X_2^2} \\ \Theta &= \tan^{-1} \left(\frac{X_2}{X_1} \right) \end{aligned} \right\} \Rightarrow \begin{aligned} X_1 &= R \cos \Theta \\ X_2 &= R \sin \Theta \end{aligned} \quad (72)$$

$$\underbrace{\mathbf{J} \begin{pmatrix} X_1, X_2 \\ R, \Theta \end{pmatrix}}_{\text{Jacobian Matrix}} = \frac{\partial (X_1, X_2)}{\partial (R, \Theta)} = \begin{vmatrix} \frac{\partial X_1}{\partial R} & \frac{\partial X_2}{\partial R} \\ \frac{\partial X_1}{\partial \Theta} & \frac{\partial X_2}{\partial \Theta} \end{vmatrix} = \begin{vmatrix} \cos \Theta & \sin \Theta \\ -R \sin \Theta & R \cos \Theta \end{vmatrix} \quad (73)$$

Joint PDF of X_1 and X_2 ,

$$\begin{aligned} f_{X_1, X_2}(x_1, x_2) &= \frac{1}{2\pi\sigma^2} \exp \left\{ -\frac{1}{2} \left[\frac{(x_1 - \mu_1)^2 + (x_2 - \mu_2)^2}{\sigma^2} \right] \right\} \\ &= f_{X_1}(x_1) \cdot f_{X_2}(x_2) \end{aligned} \quad (74)$$

Joint PDF of R and θ ,

$$\begin{aligned} f_{R, \Theta}(r, \theta) &= \mathbf{J} f_{X_1, X_2}(x_1, x_2) \Big|_{\substack{x_1 = r \cos \theta \\ x_2 = r \sin \theta}} \\ &= \frac{r}{2\pi\sigma^2} \exp \left\{ -\frac{1}{2} \left[\frac{(r \cos \theta - \mu_1)^2 + (r \sin \theta - \mu_2)^2}{\sigma^2} \right] \right\} \\ &= \frac{r}{2\pi\sigma^2} \exp \left\{ -\frac{1}{2\sigma^2} (r^2 \cos^2 \theta + \mu_1^2 - 2r\mu_1 \cos \theta + r^2 \sin^2 \theta + \mu_2^2 - 2r\mu_2 \sin \theta) \right\} \\ &= \frac{r}{2\pi\sigma^2} \exp \left\{ -\frac{1}{2\sigma^2} [r^2 + \mu_1^2 + \mu_2^2 - 2r(\mu_1 \cos \theta + \mu_2 \sin \theta)] \right\} \end{aligned} \quad (75)$$

Take,

$$\begin{aligned} 2r(\mu_1 \cos \theta + \mu_2 \sin \theta) &= 2r\sqrt{\mu_1^2 + \mu_2^2} \left(\frac{\mu_1}{\sqrt{\mu_1^2 + \mu_2^2}} \cos \theta + \frac{\mu_2}{\sqrt{\mu_1^2 + \mu_2^2}} \sin \theta \right) \\ &= 2r\sqrt{\mu_1^2 + \mu_2^2} (\cos \phi \cos \theta + \sin \phi \sin \theta) \\ &= 2r\sqrt{\mu_1^2 + \mu_2^2} \cos(\theta - \phi) \end{aligned} \quad (76)$$

where

$$\cos \phi = \frac{\mu_1}{\sqrt{\mu_1^2 + \mu_2^2}}, \quad \sin \phi = \frac{\mu_2}{\sqrt{\mu_1^2 + \mu_2^2}} \quad \text{and} \quad \tan \phi = \frac{\mu_2}{\mu_1}$$

Using eqn. (76) in (75)

$$\begin{aligned} f_{R, \Theta}(r, \theta) &= \frac{r}{2\pi\sigma^2} \exp \left\{ -\frac{1}{2\sigma^2} \left[r^2 + \mu_1^2 + \mu_2^2 - 2r\sqrt{\mu_1^2 + \mu_2^2} \cos(\theta - \phi) \right] \right\} \\ &= \frac{r}{2\pi\sigma^2} \exp \left\{ -\frac{1}{2\sigma^2} \left[r^2 + \mu_1^2 + \mu_2^2 - 2r\sqrt{\mu_1^2 + \mu_2^2} \cos \left(\theta - \tan^{-1} \frac{\mu_2}{\mu_1} \right) \right] \right\} \\ &\quad 0 \leq r < \infty, \quad -\pi \leq \theta \leq \pi \end{aligned} \quad (77)$$

If we want to find the PDF of R , then

$$\begin{aligned} f_R(r) &= \int_{-\pi}^{\pi} f_{R, \Theta}(r, \theta) d\theta \\ &= \frac{r}{\sigma^2} \exp \left\{ -\frac{(r^2 + \mu_1^2 + \mu_2^2)}{2\sigma^2} \right\} \times \left\{ \frac{1}{2\pi} \int_{-\pi}^{\pi} \exp \left\{ \frac{r\sqrt{\mu_1^2 + \mu_2^2}}{\sigma^2} \cos \left(\theta - \tan^{-1} \frac{\mu_2}{\mu_1} \right) \right\} d\theta \right\} \end{aligned}$$

we know that,

$$\underbrace{I_0(x)}_{\text{Modified Bessel function of zeroth order and first kind}} \triangleq \frac{1}{2\pi} \int_{-\pi}^{\pi} \exp \{x \cos(\theta + \phi)\} d\theta$$

$\therefore$ PDF of R ,

$$f_R(r) = \frac{r}{\sigma^2} \exp \left\{ -\frac{(r^2 + \mu_1^2 + \mu_2^2)}{2\sigma^2} \right\} I_0 \left(\frac{r}{\sigma^2} \sqrt{\mu_1^2 + \mu_2^2} \right), \quad r \geq 0 \quad (78)$$

$f_R(r)$ - marginal PDF : R has a Rician distribution when $\mu_1^2 + \mu_2^2 > 0$.

Note

- $X_1 \sim \mathcal{N}(\mu_1, \sigma^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma^2)$
 - Different mean $\mu_1 \neq \mu_2 \neq 0$
 - Same variance
 - X_1 and X_2 are independent

Then $R = \sqrt{X_1^2 + X_2^2}$ follows Rician distribution (non-zero mean).

- $X_1 \sim \mathcal{N}(0, \sigma^2)$, $X_2 \sim \mathcal{N}(0, \sigma^2)$
 - $\mu_1 = \mu_2 = 0$
 - Same variance
 - X_1 and X_2 are independent

Then $R = \sqrt{X_1^2 + X_2^2}$ follows Rayleigh distribution (zero mean).
Rayleigh PDF :

$$f_R(r) = \frac{r}{\sigma^2} \exp\left\{-\frac{r^2}{2\sigma^2}\right\}, \quad r \geq 0$$

where $I_0(0) = 1$

- $X_1 + jX_2$: Complex Gaussian random variable.

$$R = \sqrt{X_1^2 + X_2^2} \quad \Theta = \tan^{-1}\left(\frac{X_2}{X_1}\right)$$

$$f_{X_1, X_2}(x_1, x_2) \rightarrow f_{R, \Theta}(r, \theta)$$

$$f_R(r) = \int_{-\pi}^{\pi} f_{R, \Theta}(r, \theta) d\theta$$

$$f_{\Theta}(\theta) = \int_0^{\infty} f_{R, \Theta}(r, \theta) dr$$

- We know that X_1 and X_2 are independent

$$\Rightarrow f_{X_1, X_2}(x_1, x_2) = f_{X_1}(x_1) f_{X_2}(x_2)$$

$$\text{But, } f_{R, \Theta}(r, \theta) \neq f_R(r) f_{\Theta}(\theta)$$

R and Θ are not independent of each other for Rician distribution.

- If $\mu_1 = 0$ and $\mu_2 = 0$, we know that,

$$f_R(r) = \frac{r}{\sigma^2} \exp\left\{-\frac{r^2}{2\sigma^2}\right\}, \quad r \geq 0$$

$$\begin{aligned} f_{\Theta}(\theta) &= \int_0^{\infty} f_{R, \Theta}(r, \theta) dr \\ &= \int_0^{\infty} \frac{r}{2\pi\sigma^2} \exp\left\{-\frac{r^2}{2\sigma^2}\right\} dr \\ &= \frac{1}{2\pi} \int_0^{\infty} \frac{r}{\sigma^2} \exp\left\{-\frac{r^2}{2\sigma^2}\right\} dr \\ &= \frac{1}{2\pi} \int_0^{\infty} f_R(r) dr \\ &= \frac{1}{2\pi} \end{aligned} \tag{79}$$

Θ is uniformly distributed over $[-\pi, \pi]$ with PDF :

$$f_{\Theta}(\theta) = \frac{1}{2\pi}$$

$\therefore$ For $\mu_1 = \mu_2 = 0$, we have,

$$\begin{aligned} f_{R,\Theta}(r, \theta) &= \frac{r}{2\pi\sigma^2} \exp\left\{-\frac{r^2}{2\sigma^2}\right\} \\ &= \frac{1}{2\pi} \cdot \frac{r}{\sigma^2} \exp\left\{-\frac{r^2}{2\sigma^2}\right\} \\ &= f_R(r) \cdot f_\Theta(\theta) \end{aligned}$$

$\Rightarrow$ R and Θ are independent

$f_R(r)$: Rayleigh

$f_\Theta(\theta)$: Uniform

Note: X_1 and X_2 are independent

This does not hold when $\mu_1^2 + \mu_2^2 > 0$.

- The phase of a signal with a Rayleigh envelop is uniformly distributed over $[-\pi, \pi]$ and independent of the envelop.
- The phase of a signal with a Rician envelop is neither uniformly distributed over $[-\pi, \pi]$, nor independent of the envelop.
- To find $f_{R^2}(y)$:

$$F_{R^2}(y) = \Pr(R^2 \leq y) = \Pr(R \leq \sqrt{y}) = F_R(\sqrt{y})$$

$$f_{R^2}(y) = \frac{f_R(\sqrt{y})}{2\sqrt{y}}$$

We know that,

$$\begin{aligned} f_R(r) &= \frac{r}{\sigma^2} \exp\left\{-\frac{(r^2 + \mu_1^2 + \mu_2^2)}{2\sigma^2}\right\} I_0\left(\frac{r}{\sigma^2} \sqrt{\mu_1^2 + \mu_2^2}\right), \quad r \geq 0 \\ f_{R^2}(y) &= \frac{1}{2\sqrt{y}} \cdot \frac{\sqrt{y}}{\sigma^2} \exp\left\{-\frac{(y + \mu_1^2 + \mu_2^2)}{2\sigma^2}\right\} I_0\left(\frac{\sqrt{y}(\mu_1^2 + \mu_2^2)}{\sigma^2}\right) \\ &= \frac{1}{2\sigma^2} \exp\left\{-\frac{(y + \mu_1^2 + \mu_2^2)}{2\sigma^2}\right\} I_0\left(\frac{\sqrt{y}(\mu_1^2 + \mu_2^2)}{\sigma^2}\right), \quad y \geq 0 \end{aligned} \tag{80}$$

R^2 has a non central Chi-squared distribution with degrees of freedom = 2 ($\chi^2(2)$) when $\mu_1^2 + \mu_2^2 > 0$.

- When $\mu_1 = \mu_2 = 0$, we get,

$$f_{R^2}(y) = \frac{1}{2\sigma^2} \exp\left\{-\frac{y}{2\sigma^2}\right\}, \quad y \geq 0 \tag{81}$$

R^2 has a central Chi-squared distribution with degrees of freedom = 2 ($\chi^2(2)$) when $\mu_1 = \mu_2 = 0$. Also called, R^2 has an exponential distribution when $\mu_1 = \mu_2 = 0$.